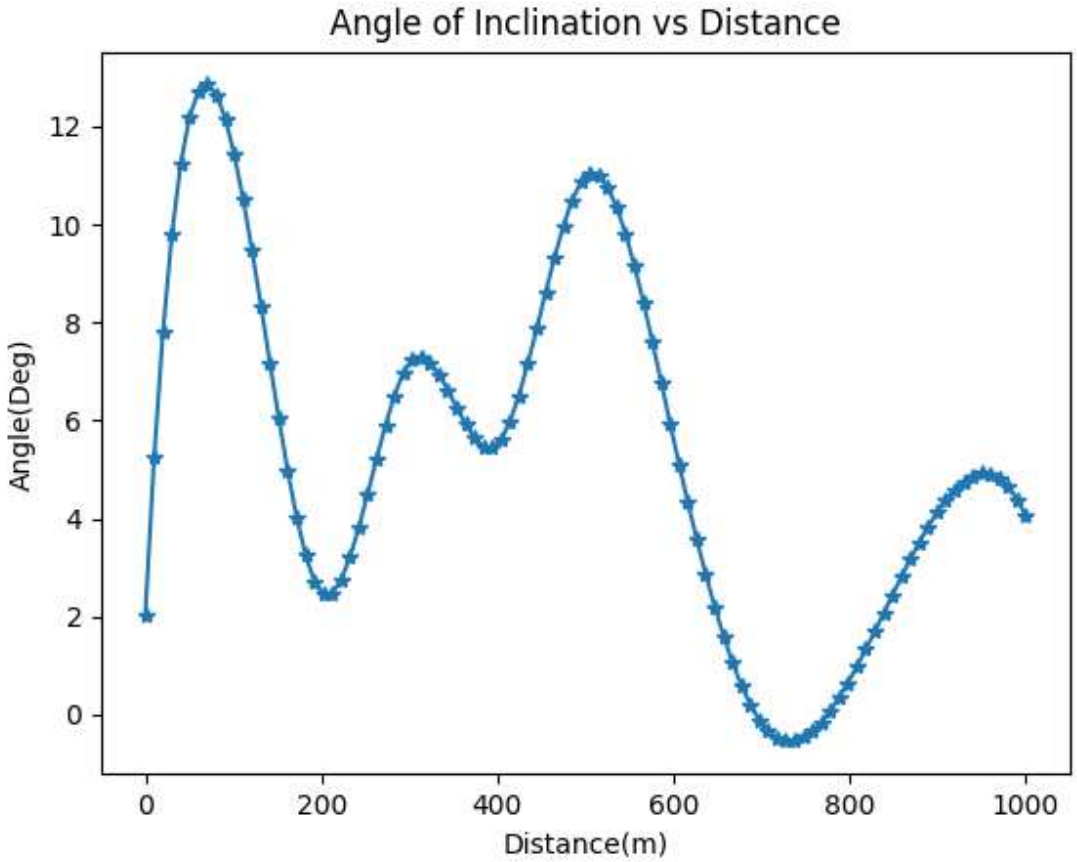
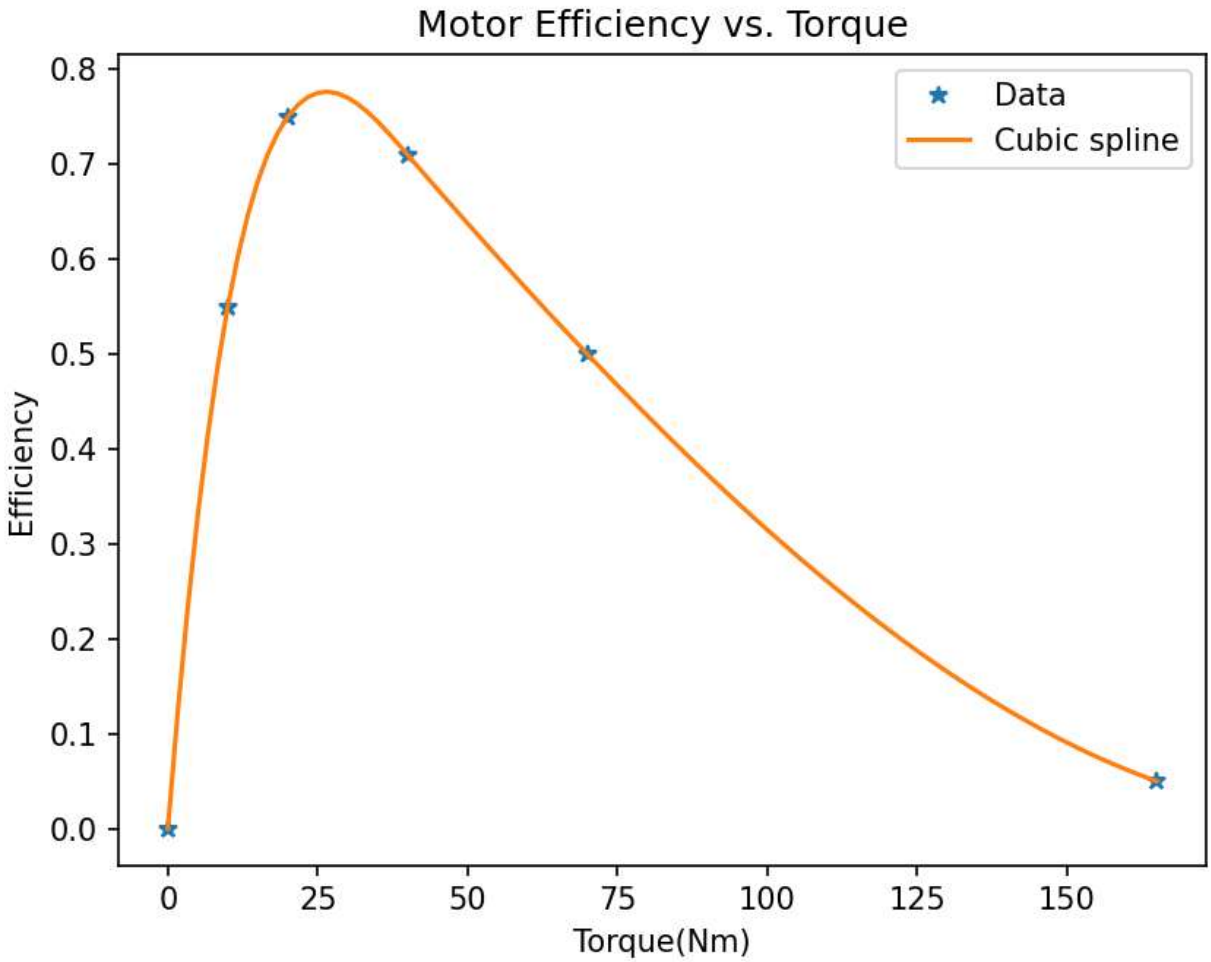
Task 2:



This figure shows the angle of the terrain at various distances. The curve produced is made by fitting a spline along the given points. The spline, as opposed to other methods such as linear line segments, produces a smooth curve. All the original data points fall on this line, for example, we can see (200, 2.478) and (400, 5.511).

Task 5:



This graph represents a ***smooth*** relationship between torque and motor efficiency. Using 100 points evenly spaced between the maximum torque value of 165Nm and the minimum torque value of 0 Nm, we used a cubic spline to interpolate between the provided data and graphically plot as shown above. The spline goes through each given point which is represented with a "*". It shows that the motor has a max efficiency of approximately 75.75% at around 30Nm with a cubic decline afterwards. This differs from Figure 2 (the graph shown below) because the max efficiency is slighter higher, and the most efficient torque value is more than 100 times greater than figure 2's most efficient torque value. However, the relationship trends are very similar and exhibited the same behavior.

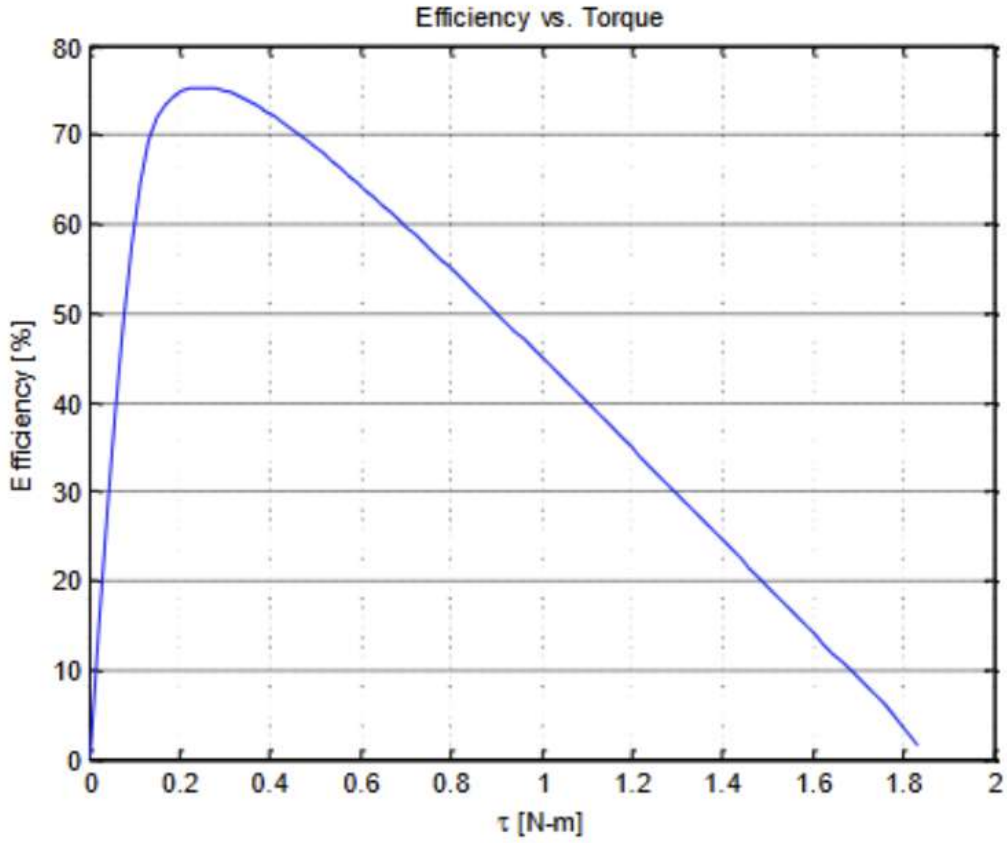
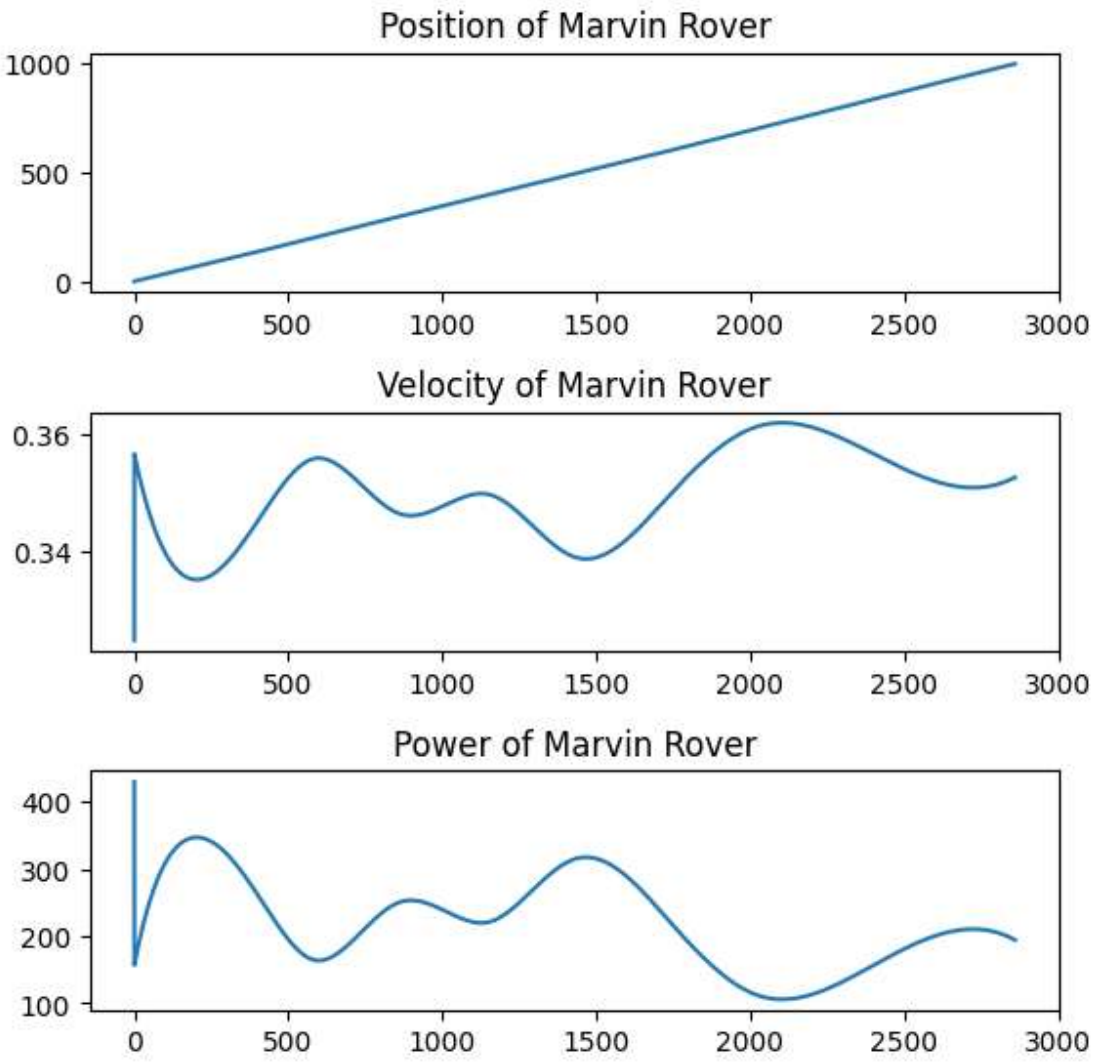


Figure 2: Efficiency curve for a typical DC motor.

Task 8:



The graph of position appears linear despite velocity being nonlinear due to the fact that Marvin maintains a relatively constant velocity. The velocity stays mostly in the range of 0.34 - 0.36 m/s, meaning that despite the graph seeming very nonlinear, the actual fluctuation is small. The power and velocity graphs are inversely proportional. This is because both low velocity and high power are caused by Marvin needing to overcome difficult terrain. This can be seen when looking at the experiment visualization graph, as power matches up exactly to the curves seen there. The velocity graph is smooth due to the motor needing to apply torque/acceleration in order to move the rover which cannot cause a sudden change in velocity. The integration method was also swapped from RK45 to BDF after testing following being told about the method in class, which smoothed the graph more than it already was. This implies

that the system may exhibit some stiffness, as BDF handles stiffer systems better than explicit single step methods.

Task 9:

If the rover is equipped with a 0.9072E6 J Lithium Iron Phosphate battery the rover cannot complete the case defined by experiment 1. This is due to the fact that the energy consumed by the trip, which was found to be 1.1282E6 Joules by applying the battenergy function to the time and velocity values found with simulate_rover, is more than the available energy.